

Examination	:	End-Semester Examination – May/June 2026
Name of the Course	:	B.Tech. (IT & Mathematical Innovations)
Name of the Paper	:	Information Security
Unique Paper Code	:	3122613602
Semester	:	VI
Duration	:	3 hours
Maximum Marks	:	90

Instruction to students:

Attempt any five questions out of seven. Question No. 1 is compulsory. All questions carry equal marks. Scientific calculators without storage memory are allowed.

- Q.1 Answer the following questions. (6*3)
 - a) Differentiate between the terms *privacy* and *security* with suitable example.
 - b) While securing multimedia information what are the key issues to address?
 - c) Compare working of monoalphabetic and polyalphabetic ciphers.
- Q.2 a) An academic institution wants to secure its campus from unwanted/ unauthorised network access from outside the institution while internal accessibility is strengthened. This aims to protect the important assets/ Labs from unauthorised access or data breach. List the steps to be followed to achieve this goal. Also suggest the devices/ software and their types (if any), configuration and place of installation in entire network to be used in this exercise. (9+9)
- b) What is a VPN? Explain its working and mention its capabilities.
- Q.3 Two parties A and B need to share some documents over an unprotected network. Initially to start the communication both parties require to setup a key agreement. At first, the key is not shared and both of them will agree on a certain key before communication. Suggest one algorithm in this scenario for key agreement with an example. Assume necessary values if required. (18)
- Q.4 What is a Public key cryptosystem? How it is different from private key cryptosystem? Encrypt the text "*privacy*" using RSA algorithm. Consider ASCII encoding scheme for characters. Also take the two prime numbers less than 19 if required. (18)
- X Q.5 Compare CFB, OFB and CTR modes of AES algorithm. Explain the advantages and disadvantages of each mode. List most suitable applications for each mode. (18)

XQ.6 a) What is X.509 certificate? What are its contents? Explain its role in secure (9+9) communication.

b) Encrypt the text "MONARCHYENDED" using Vigenère cipher with a key "wearedone". Explain the steps involved.

Q.7 Write short notes on any three of the following. (6*3)

a) Secure Electronic Transactions

b) SHA

c) Dictionary Attacks

d) Intrusion Prevention System
